

Timothy Do's Official CV

An A.I. Machine Vision Enthusiast & DSP Techie to Make the World A Better Place

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EDUCATION

University of California Irvine, Irvine, CA — B.S. Electrical Engineering

September 2019 - June 2023 (Expected)

- 3rd Year Undergraduate, Senior Standing
- Overall GPA: 3.90, Major GPA: 3.93
- **Specialization:** Digital Signal Processing
- **Research Interests:** Digital Signal Processing with Artificial Intelligence, Image Enhancements, Motion Interpolation, and Pixel Manipulation to make media more enjoyable.
- **Relevant Coursework:**
 - EECS 10: An Introduction to C Programming, learned about compilers, data structures, and linux commands (Fall 2019)
 - EECS 31L: Digital Logic Lab, experience with VHDL, applying knowledge from EECS 31 into applications. (Spring 2020)
 - EECS 22: Advanced C Programming, learned about advanced data structures, program decomposition (Winter 2021)
 - EECS 22L: Introduction to Software Engineering, worked in a team of 4 (TTL22s) to develop a network chess application in C with a git repo with version control (Spring 2021)
 - EECS 50: Discrete-Time Signals and Systems, learned about LTI Systems, Z-Transforms, the DTFT, and FFT (Spring 2021)
 - EECS 70B: Network Analysis II, learned about Two-Port Circuits, Laplace Transforms with Transient Circuits, and Frequency Response (Spring 2021)
- **Major Coursework** for Fall 2021: EECS 170A (Semiconductors), EECS 152A (Digital Signal Processing), and EECS 145 (EE Analysis)
- **Awards:** Dean's Honor List (GPA 3.5+) for 6 Quarters (Fall 2019, Winter 2020, Spring 2020, Fall 2020, Winter 2021, Spring 2021)

Evergreen Valley High School, San Jose, CA — Diploma

August 2015 - May 2019

- GPA: Unweighted: 3.8, Weighted: 4.0
- **Relevant Coursework:** AP Physics C: Mechanics (4), AP Computer Science A (4), AP Statistics (4), AP Calculus AB (5), AP US History (4), Math 072 (Calc II) @ Evergreen Valley College
- **Awards:** EVHS Band Scholar, Magna Cum Laude, Golden State Diploma, Bilingual Certification in Vietnamese, AP Scholar with Distinction

SKILLS

Musicianship:

Played the Euphonium in an ensemble setting

Teaching:

Created teaching plans for music tutoring & lead sections of the band during rehearsal

Planning:

Planned and delegated the blueprint design cabinets for my Eagle Scout Project

Web Programming:

Created Websites with HTML, CSS, and Javascript

Compiler / Object-Oriented Programming:

Experience with C, Java, and Python libraries

GPIO Programming:

Used MicroPython or Ino to control circuits digitally with Arduino or Raspberry Pi.

Photo/Video Editing:

Edited Videos/Photos as Final Cut Pro, Photoshop, & Premiere Pro.

Computers:

Assembled/Debugged Desktop machines with OS imaging.

CUDA:

Programmed GPUs to do Matrix Multiplications and DSP

ACTIVITIES

Member of UCI IEEE Club

Member of UCI Micromouse

Former 3rd Chair Euphonium Player for the 2019 Santa Clara County Honor Band

Former Euphonist for EVYO (Evergreen Valley Youth Orchestra)

Former Teacher's Assistant for Most Holy Trinity Parish

EXPERIENCE

Most Holy Trinity Parish, San Jose, CA — *Teacher's Assistant*

September 2013 - May 2019

- Helped young students in their religious studies in Catholicism and their skills in Vietnamese.
- Overviewed Teacher's Plan and curated lessons for students to learn by presenting premade powerpoints.
- During study hall, go around the class and help the students with any questions 1-1.

Boy Scouts of America Troop 610, San Jose, CA — *Eagle Scout*

September 2013 - September 2019

- Mentored youth between 12-17 on the principles of scouting
- Awarded the Bronze Palm: Earning 5+ additional Merit Badges over the 21 required for Eagle Scout.
- **Eagle Scout Project:**
 - Constructing 2 Food Cabinets for the Milpitas Food Bank
 - Required detailed planning, sketches, and team management
- **Troop Roles:**
 - Bugler: Played the trumpet/Euphonium during troop outings
 - ASPL: Managed troop as a whole with their daily activities, planned outings for the troop to goto

Evergreen Music Mentors, San Jose, CA — *Founding President*

September 2018 - Present

- Lead music tutoring sessions for younger middle school students
- Managed private tutoring sessions with employees and clients
- Managed financial transactions for the company
- Integrated software (Cybernotes) as a video calling platform

Environmental Injustice , Irvine, CA — *Collaborator*

June 2020 - Present

- Admin for the discord server, managing working groups
- Managed Instagram account, creating captions/visuals using Canva.
- Collaborated with 11 International Team Members remotely to research on chemical disasters, pollution, and climate change
- Developed techniques such as stakeholder analysis, using Civic Data Tools such as CalEnviroScreen , and citing in Chicago format.

UC Irvine D.C.E., Irvine, CA - *I.T. General Assistant*

August 2021 - Present

- Helped Set-Up Desktops, Docks, Webcams, and Computers for over 300+ Employees in UCI's Division of Continuing Education
- Worked with Network Administrators by managing IPs, Subnet Masks, and DNS servers of DCE users to access local services in the Network
- Diagnosed various problems users are having (i.e. not connected to the VPN, Monitor is not displaying) , and finding solutions to fixing them by employing IT tactics (pinging, searching local devices, etc.)

Boy Scouts of America, San Jose, CA - *Merit Badge Counselor*

July 2021 - Present

- Offered to teach three Merit Badge Courses: Programming, Music, and Bugling.
- Taught a Group of 10-12 Scouts the History of Programming, Debugging Practices, IP(s), and assisted them in any programming questions for their Merit Badge Requirements
- Managed the Scout's Requirements by creating Google Forms / Google Sheets for easy submission of their programming work online.
- Outreached to Boy Scout Troop 610 with weekly updates to encourage more scouts to sign up (inclusivity)

PUBLICATIONS

Tahis Alcantar; **Timothy Do**; April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. [Fast Disaster Case Study: Santa Fe Springs](#). UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California Irvine.

Tahis Alcantar; **Timothy Do**; April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. [Slow Disaster Case Study: Wilmington](#). UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California Irvine.

Tahis Alcantar; **Timothy Do**; April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. [Combo Disaster Case Study: San Bernardino County](#). UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California Irvine

LANGUAGES

English (Fluent), **Vietnamese** (Fluent), & **Spanish** (Proficient)

HACKATHONS

Cisco Webex Hackathon 2021

Gratition: An Easing Mental Health Web App (with Google Voice Synthesis)

Languages: HTML/CSS/JS

ZotHacks 2021

Anteatings: A Fun JS-Based Antsmashing Game on the Web

Languages: HTML/CSS/JS

PROJECTS

[TTLZNetChess](#) - *An Advanced Client-Based Chess Application*

March 2021 - June 2021

- Managed a team of four developers into develop a C-Based Chess Application for EECS22L (Intro to Software Engineering)
- Implemented Terminal Graphics (Colors) using Escape Codes
- Implemented AIs in Chess using Minimax Algorithm
- Networking Support using standard sockets and FCFS Protocol

[RefreshVideos](#) — *A High Refresh Rate Video Player*

April 2020 - Present

- **Languages Used:** HTML / CSS / Javascript
- Used Canvas to play videos at high frame rates and apply effects
- Featuring filters like Edge Detection, Aging, etc.

[pdDSP](#) — *Audio-Processing using Pure-Data*

March 2020

- **Languages Used:** Pure Data
- Used Pure Data to mix audio and designed audio filters (e.g. Low Pass Filter, Band Pass Filter)

[PiFmMorse](#) — *A Morse Code FM Transmitter for the Raspberry Pi*

September 2020

- **Languages Used:** Python
- An Program to transmit Morse Code over FM radio via GPIO 4
- Learned to concatenate .wav files via wave library
- Learned to call linux commands using os library

[RPiMorse](#) — *A Morse Code LED Circuit for the Raspberry Pi*

September 2020

- **Languages Used:** Python
- An Program to flash an LED in Morse Code given a circuit
- Learned GPIO Pins Wiring and its respective python library

[Timduino Tutorials](#) — *Experiments with the Arduino Uno*

September 2020

- **Languages Used:** C
- Learned functions to set voltages (pinMode(), digitalWrite())
- Wired Circuits and pins to apply Arduino Programs

[SCU Complex Fires](#) — *A Website Providing Wildfire Resources*

August 2020 - September 2020

- **Languages Used:** HTML / CSS/ Javascript
- Provided Wildfire resources for research, shelter, and news
- Worked with a team of 4 and remotely collaborated through Github
- Features an Energy-Efficient Dark Mode.